

Supplementary Information

AI_DNA_ANALYZER: exploratory neural sequence-caller branch

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This Supplementary Information documents the exploratory neural branch referred to in the Introduction of the main manuscript (“... apart from one weak positive (two of three seeds) on a side branch (Mamba-style models [16]; Supplementary Information)”). No part of this branch is used by V1, V2 or V2.x, and nothing here changes any result of the main manuscript. Every number below is copied from a project record named next to it; paths are relative to the repository root. *Devlog* *n* means `History/n_DEVLOG.md`. Reference numbers refer to the main manuscript’s reference list.

S1. Origin: a neural caller on synthetic reads (August 4–10, 2026)

The project began as a per-base token classifier (`model.py`) predicting Normal/SNP/Insertion/Deletion from a read sequence, the aligned reference, base quality and depth. The source describes it as a reference-guided bidirectional “Mamba” variant caller, but its own docstring states that the block “is **not** a selective state-space model”: it is a causal gated convolutional approximation in stock PyTorch, with a receptive field bounded by the convolution kernel. “Mamba-style” in the main text refers to this architecture and its later variants, not to the selective state-space model of [16].

Training and evaluation data were synthetic reads (`generate_mock_bam.py`) carrying variants from `generate_mock_vcf.py`, placed on a slice of the real chr21 reference. Real HG002 reads were not used in this phase, so it produced no measurement of real-data accuracy. The records of this phase (devlogs 3–5, `History/DEVLOG_2026-08-08.md`) mainly document data and evaluation defects:

- The mock BAM spliced only SNPs, so the indel classes could not be learned, and the vocabulary lacked a gap token (devlog 3).
- A fixed 0.5 decision threshold made one checkpoint score macro-F1 0.0884; the same checkpoint scored 0.9095 at threshold 0.9 (`History/DEVLOG_2026-08-08.md`). The apparent “collapse” was a consequence of the evaluation rule.
- The mock-VCF generator drew the REF allele at random, so 81 of 374 SNP labels had no read evidence, which capped recall at 0.7834 (devlogs 3–4). After the fix the devlog declared all earlier metrics non-comparable.

Per-seed numerical records for the synthetic-read runs (devlogs 3 and 5 mention two seeds) are not retained in a form that can be tabulated here, and none are reported.

S2. Neural stages on real reads (August 11–12, 2026)

From devlog 6 onward, all experiments used real GIAB HG002 reads on chr21 (GRCh38, NIST v4.2.1 truth) at about 15×. The first test region was chr21:30.0–30.47 Mb (455,371 test loci, 549 SNPs; devlogs 9–10). Each neural variant was compared with a statistical baseline that used the same data:

Neural variant	Result (test split)	Source
Residual Mamba initialised at the binomial LLR	NULL. The validation-selected model is identical to the binomial caller (521 TP, 1 FP, 28 FN; F1 0.9729). A forced-training arm gave +0.0038 F1, but it did not replicate across seeds and was matched by moving the binomial threshold from 10.5 to 9.5 (F1 0.9778).	devlog 6 §5
Read-level Mamba (per-read attributes)	NEGATIVE. 499 TP / 65 FP relative to the binomial caller. Decoupling each read’s attributes from its base <i>improved</i> F1 by 0.0427.	devlog 7 §§11–12
Neural residual above the Poisson-binomial (PB) caller	NULL. Calls identical to PB for every checkpoint (539 TP, 0 FP, 10 FN; F1 0.9908; $\Delta F1 = 0.0000$).	devlog 9 §6

The positive result of this period was statistical rather than neural: the quality-aware Poisson-binomial LLR (devlog 8) is the PB stage used in V1.

S3. The Mamba-style side branch: PB-residual on a powered benchmark

Devlog 9 found that its test region was too small to show a difference (10 PB errors in total). The PB-residual model was therefore re-evaluated on a larger split, on side branch `worktree-30x-pb-residual` (commit 6c31876, 2026-08-12). The data were HG002 chr21:32–40 Mb at about 15 $\times$, divided into 1-Mb blocks: train blocks 0 and 4 (1.89 Mb, 2,544 SNPs), validation blocks 1 and 5 (1.91 Mb, 2,410 SNPs) and test blocks 2, 3, 6 and 7 (3.81 Mb, 5,704 SNPs). This is the same split described under “Provenance of the constants” in the main manuscript. Each arm’s decision threshold was chosen on validation, and the test blocks were scored once. Three training seeds were used: A = 20260811, B = 424242 and C = 13371337. The paired bootstrap against PB used 10,000 resamples with seed 20260812.

Method (test, 3.81 Mb)	Precision	Recall	F1	TP	FP	FN
Binomial	0.9747	0.9385	0.9562	5,353	139	351
Poisson-binomial (PB)	0.9899	0.9623	0.9759	5,489	56	215
PB-residual, seed A	0.9921	0.9627	0.9771	5,491	44	213
PB-residual, seed B	0.9933	0.9628	0.9778	5,492	37	212
PB-residual, seed C	0.9906	0.9623	0.9763	5,489	52	215
Control: shuffled PB prior	0.5518	0.9862	0.7077	5,625	4,568	79

Paired vs PB	$\Delta F1$	95 % CI	ΔFP	ΔFN
Seed A	+0.0012	[+0.0001, +0.0024]	−12	−2
Seed B	+0.0019	[+0.0009, +0.0030]	−19	−3
Seed C	+0.0003	[−0.0007, +0.0015]	−4	0

Source: `results/big15x/test_results_3seed.json` on branch `worktree-30x-pb-residual`; narrative in devlog 10.

Two of the three seeds have intervals that exclude zero. The third was run as an unconditional replication check after A and B had been scored, and its interval includes zero. The branch record

therefore gives the verdict **WEAK POSITIVE**. Two controls from the same record are relevant. The best additive constant on the PB score, fitted on validation, was 0.0, so the gain is not a threshold shift. With the PB prior shuffled between loci, the model fell to F1 0.7077. In seed B, PB had 56 false positives. The model removed 19 of them and added none.

S4. The side branch inside the cascade

The trained side-branch checkpoints were then tested as a third stage (PB $\rightarrow$ Mamba) on chr21:31–32 Mb, a region held out from their training (devlog 12). The pre-registered verdicts were as follows. **Stage 2 (PB $\rightarrow$ Mamba): NEGATIVE.** Two of the three checkpoints were significantly harmful ($\Delta F1$ -0.018 to -0.073), and the third was null. **Stage 3 (full cascade):** the best case was $\Delta F1$ $+0.0002$ with an interval straddling zero, at a window amplification of 31.7 – $61.0\times$. The two-stage binomial $\rightarrow$ router $\rightarrow$ PB cascade was the Pareto-optimal arm. The same devlog reports the PB depth-clipping defect above $48\times$ that was fixed in devlog 13. This Supplement does not re-examine whether that defect affected the Stage-2/3 numbers. They are reported as recorded.

S5. Why the neural branch is not part of V1, V2 or V2.x

1. On real reads, every neural stage apart from the side branch was null or negative against a statistical baseline (S2).
2. The side-branch gain was small ($+0.0003$ to $+0.0019$ F1), replicated in two of three seeds only, and did not survive integration into the cascade on held-out data (S3, S4).
3. The side branch depends on the PB prior (the shuffled-prior control fell to F1 0.7077), so it could only be added on top of the statistical caller, not replace it.

V1 was therefore built from the statistical stages alone. `cascade.py` still contains a dormant Mamba stage, but it is not used by any reported result. The supervised component of V2.x is a logistic regression and is unrelated to this branch (main manuscript).

S6. Records used and records not available

Used: `model.py`; `History/3_DEVLOG.md`–`12_DEVLOG.md`; `History/DEVLOG_2026-08-08.md`; `research/RESEARCH_TIMELINE` (E1–E11); `research/CLAIM_EVIDENCE_MAP.csv`; branch `worktree-30x-pb-residual` (commit 6c31876 and its `results/big15x/*.json`).

Not available: Detailed numerical records for the synthetic-read phase (per-seed, per-epoch metrics) are not retained in the current repository in tabulable form. No run with more than three seeds and no evaluation region larger than 3.81 Mb was carried out for the side branch. The follow-ups recommended in devlog 10 (30–50 Mb, 5–10 seeds) were not run.